STUDENT NUMBER:

APSC 171 - Term Test 2
Tuesday November 7, 2023

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- This is a 2 hour test.
- No textbooks, notes, or aids other than your calculator (Casio 991 series) are allowed.
- For full marks, you must show all your work and explain how you arrived at your answers, unless explicitly told to do otherwise.
- Wherever appropriate, **include units in your answers**.
- When drawing graphs, **add labels and scales on all axes**.
- Correct final answers with incorrect support may be given no credit.
- While waiting to start, fill in your name and student number above.
- If you need more room, there are blank pages at the end of the test. **If you use these pages, you must provide clear directions on the original page**, e.g. "Continued on page 10".
- Only answers written in this booklet will be graded; all scrap paper from the test will be discarded.

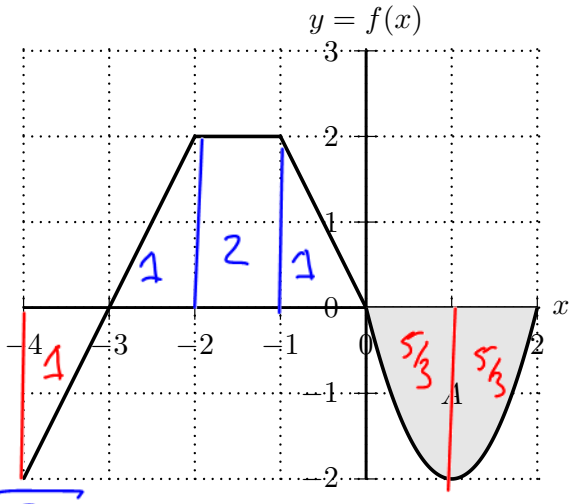
Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Total
7	12	7	7	7	8	6	6	60

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1. Determine the value of the following integrals, based on the graph of $f(x)$ shown on the right hand side.
   Note: the shaded region A has area $\frac{10}{3}$ square units.
   Either show your calculations, or add appropriate annotations to the graph to support your answers.

(a) $\int_{-3}^{0} f(x)\,dx$ [/2]

$= 4$



(b) $\int_{-4}^{0} f(x)\,dx$ [/1]

$= \int_{-4}^{-3} f(x)\,dx + \int_{-3}^{0} f(x)\,dx = -1 + 4 = 3$

(c) $\int_{-4}^{1} f(x)\,dx$ [/1]

$= \int_{-4}^{-3} f(x)\,dx + \int_{-3}^{0} f(x)\,dx + \frac{1}{2}\int_{0}^{2} f(x)\,dx$

$= -1 + 4 - \frac{5}{3} = \frac{4}{3} = 1.3\dot{3}$

(d) Evaluate the definite integral $\int_{-3}^{-1} \frac{1}{x}\,dx$. (Unrelated to the previous parts.) [/3]

$\int_{-3}^{-1} \frac{1}{x}\,dx = \ln|x| \Big|_{-3}^{-1} = \ln|-1| - \ln|-3|$

$= \ln(1) - \ln(3) = -\ln(3)$

or $\approx 0 - 1.0986$

$= -1.0986$

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2. Evaluate the following indefinite integrals. As always, show your work.

(a) $\int x^2 \cos(4x^3)\, dx$ [/4]

$I_1 =$ By substitution

Let $w = 4x^3 \Rightarrow \frac{dw}{dx} = 12x^2 \Rightarrow \frac{1}{12x^2}\, dw = dx$

So $I_1 = \int x^2 \cos(w)\left(\frac{1}{12x^2}\, dw\right) = \frac{1}{12}\int \cos(w)\, dw$

$= \frac{1}{12}\sin(w) + C$

$= \frac{1}{12}\sin(4x^3) + C$

(b) $\int \tan(x)\, dx$ [/4]

Recall: $\tan(x) = \frac{\sin(x)}{\cos(x)}$

$= \int \frac{\sin(x)}{\cos(x)}\, dx$ → By substitution, let $w = \cos(x)$

$\Rightarrow dw/dx = -\sin(x)$

$= \int \frac{\sin(x)}{w}\left(\frac{-1}{\sin(x)}\, dw\right)$ ← or $\frac{-1}{\sin(x)}\, dw = dx$

$= -\int \frac{1}{w}\, dw = -\ln|w| + C = -\ln|\cos(x)| + C$

(c) $\int \frac{3x^2 - \sqrt{x}}{x^{1/3}}\, dx$ [/4]

Rewrite first

$= \int 3x^{5/3} - x^{1/6}\, dx = 3\frac{x^{8/3}}{8/3} - \frac{x^{7/6}}{7/6} + C$

$= \frac{9}{8}x^{8/3} - \frac{6}{7}x^{7/6} + C$

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3. Consider the integral $\underbrace{\int \frac{\ln(x)}{\sqrt{x}}\,dx}_{I}$. [/6]

$= \int \ln(x) \cdot x^{-1/2}\,dx$

(a) Evaluate the indefinite integral.

By parts. Let $u = \ln(x)$ $\quad \int dv = \int x^{-1/2}\,dx$

$\downarrow d/dx$

$du = \frac{1}{x}\,dx \quad \leftarrow \quad v = 2x^{1/2}$

so $I = \ln(x)(2x^{1/2}) - \int 2x^{1/2} \cdot \frac{1}{x}\,dx$

(tidy) $= 2\ln(x)\sqrt{x} - 2\int x^{-1/2}\,dx$

(integrate) $= 2\ln(x)\sqrt{x} - 2(2x^{1/2}) + C$

$= 2\ln(x)\sqrt{x} - 4\sqrt{x} + C$

(b) Determine by differentiating whether or not your answer to part (a) is correct. State your conclusion clearly for the grader. [/1]

$\frac{d}{dx}\left(2\ln(x)\sqrt{x} - 4\sqrt{x} + C\right)$

$= \left(2\left(\frac{1}{x}\right)\sqrt{x} + 2\ln(x)\cdot\frac{1}{2}x^{-1/2}\right) - 4\cdot\frac{1}{2}x^{-1/2}$

$= \frac{2}{\sqrt{x}} + \frac{\ln(x)}{\sqrt{x}} - \frac{2}{\sqrt{x}} = \frac{\ln(x)}{\sqrt{x}}.$

This equals the original integrand, so our answer to part (b) is correct.

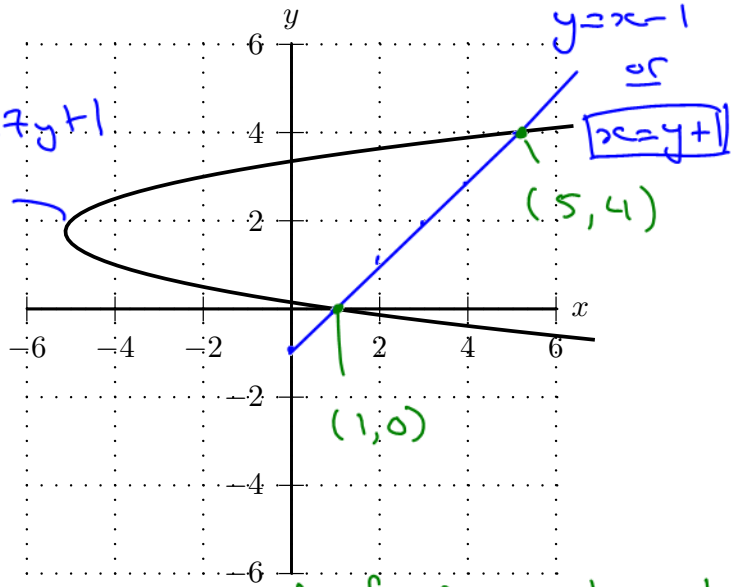
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4. Consider the graph shown at right, defined by $x = 2y^2 - 7y + 1$.

(a) On the same axes to the right, add the graph of $y = x - 1$. [/1]

slope 1, y-intercept −1

$x = 2y^2 - 7y + 1$



(b) Write an integral that represents the finite area enclosed between the graphs of $y = x - 1$ and $x = 2y^2 - 7y + 1$. [/3]

limits determined from sketch

$$\text{area} = \int_{y=0}^{y=4} x_{high} - x_{low}\ dx$$

$$= \int_0^4 (y+1) - (2y^2 - 7y + 1)\ dy$$

tidying $$= \int_0^4 -2y^2 + 8y\ dy$$

optional: finding intercepts

$x_{line} = x_{parabola}$

$y + 1 = 2y^2 - 7y + 1$

$0 = 2y^2 - 8y = 2y(y-4)$

$y = 0,\ y = 4$

(c) Evaluate the integral from part (b). [/3]

$$\int_0^4 -2y^2 + 8y\ dy = \left[-\frac{2y^3}{3} + \frac{8y^2}{2} \right]_0^4$$

$$= \left(-\frac{2}{3}(64) + 4(16) \right) - (0 + 0)$$

$$= \frac{64}{3} = 21.\dot{3} \text{ square units}$$

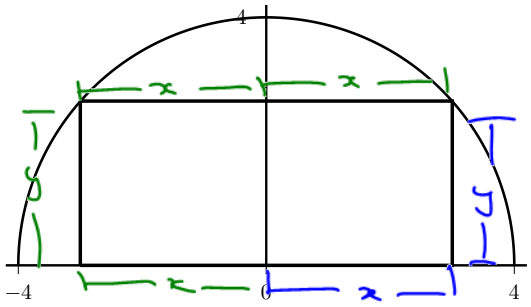
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5. Consider all possible rectangles that can fit in the semi-circle of radius 4 shown below, given that:

   - the bottom of the rectangle must lie on the x axis, and
   - the upper corners of the rectangle must lie on the semi-circle.

   Assume all dimensions are in meters.

   Find the dimensions of the rectangle of this type with the **longest perimeter**. Record your final numeric values in the entries at the bottom of the page.
   You do **not** need to show that your answer is a local maximum for the perimeter. [/7]



Perimeter $= 4x + 2y$

y is constrained by being on the semi-circle $x^2 + y^2 = 16$, or $\boxed{y = \sqrt{16 - x^2}}$

$P(x) = 4x + 2\sqrt{16 - x^2}$

Find critical points of $P(x)$:

$P'(x) = 4 + \not{2}\left(\tfrac{1}{\not{2}}(16 - x^2)^{-1/2} \cdot (-2x)\right)$

$= 4 - \dfrac{2x}{\sqrt{16 - x^2}}$ Set $= 0$ $\rightarrow$ $2 = \dfrac{x}{\sqrt{16 - x^2}}$

$x = 2\sqrt{16 - x^2}$ square both sides

$x^2 = 4(16 - x^2)$

$5x^2 = 64$

$x = \pm\sqrt{\frac{64}{5}}$ $\rightarrow$ only need positive value $x \approx 3.578$ $x = 3.58$ m

$\Rightarrow y = \sqrt{16 - x^2} \approx 1.789$

$\Rightarrow$ optimal $P = 4x + 2y \approx 17.89$

optimal width = $2x = 7.16$ m

optimal height = 1.79 m

corresponding perimeter = 17.9 m

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6. An engineering team is conducting tests on a newly designed gearbox for industrial machinery. As the gearbox operates, it records the rotational velocity of its main shaft in revolutions per minute (RPM).

The data recorded over a 10 second time interval is shown in the tables below:

Time (seconds)	0	2	4	6	8	10
Rotational Velocity (RPM)	600	620	640	580	590	615

Note: must convert RPM to RPS, or seconds to minutes.

RPS = $\frac{1}{60}$ RPM | 10 | $10.\dot{3}$ | $10.\dot{6}$ | $9.\dot{6}$ | $9.8\dot{3}$ | 10.25

(a) Use a 5-interval left sum estimate of the number of rotations of the gearbox over the 10 seconds recorded in the table. Include units in your answer. [/3]

$$LEFT(5) = \underset{(s)}{2}\left[10 + 10.\dot{3} + 10.\dot{6} + 9.\dot{6} + 9.8\dot{3}\right]_{RPS}$$

$$= 2(50.5) = 101 \text{ revolutions}$$

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or convert time interval units

$$LEFT(5) = \underset{(min)}{\frac{2}{60}}\left(600 + 620 + 640 + 580 + 590\right)_{RPM}$$

$$= 101 \text{ revolutions}$$

(b) Use a 5-interval trapezoidal estimate of the number of rotations of the gearbox over the 10 seconds recorded in the table. Include units in your answer. [/3]

$$TRAP(5) = \underset{(s)}{\frac{2}{2}}\left(10 + 2\cdot 10.\dot{3} + 2\cdot 10.\dot{6} + 2\cdot 9.\dot{6} + 2\cdot 9.8\dot{3} + 10.25\right)_{RPS}$$

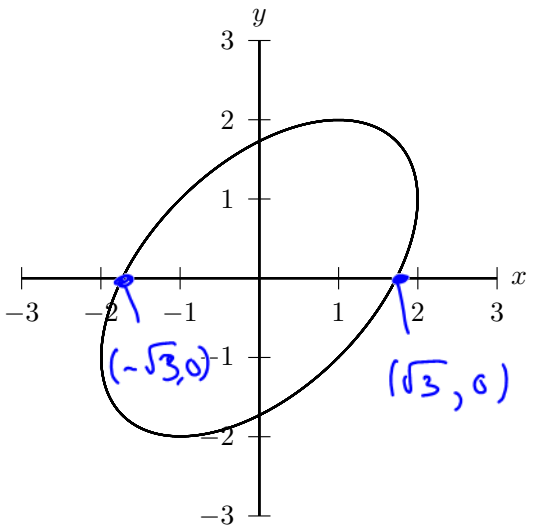
$$= \frac{405}{4} = 101.25 \text{ revolutions}$$

(c) Is it possible to use a 5-interval mid-point estimate for the number of rotations over the 10 seconds recorded? If so, perform the calculations; if not, explain why not. [/2]

No. For MID(5), we would need data from $t = 1, 3, 5, 7$ and 9s, and that is not available.

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7. Consider the relation $x^2 - xy + y^2 = 3$, which defines the ellipse drawn on right.



(a) Find a formula for $\frac{dy}{dx}$ for points on this ellipse. [/3]

Take $\frac{d}{dx}$ of both sides:

$$\frac{d}{dx}\left(x^2 - xy + y^2\right) = \frac{d}{dx}(3)$$

$$2x - y - x\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$$

Solve for $\frac{dy}{dx}$:

$$\frac{dy}{dx}(-x + 2y) = -2x + y \rightarrow \boxed{\frac{dy}{dx} = \frac{y - 2x}{2y - x}}$$

(b) Find the coordinates of the two points where the ellipse crosses the x axis, and indicate them on the graph. [/1]

x axis $\Rightarrow$ $y = 0$ $\rightarrow$ $x^2 - x(0) + 0^2 = 3$

$x^2 = 3$

$x = \pm\sqrt{3}$

Intercepts at $(\sqrt{3},0)$ and $(-\sqrt{3},0)$

(c) Use an appropriate calculation to show that at the two points found in part (b), the tangent lines to the ellipse have the same slope. (b). [/2]

At $(\sqrt{3},0)$, $\frac{dy}{dx} = \frac{0 - 2\sqrt{3}}{2\cdot 0 - (\sqrt{3})} = \frac{-2\sqrt{3}}{-\sqrt{3}} = 2$.

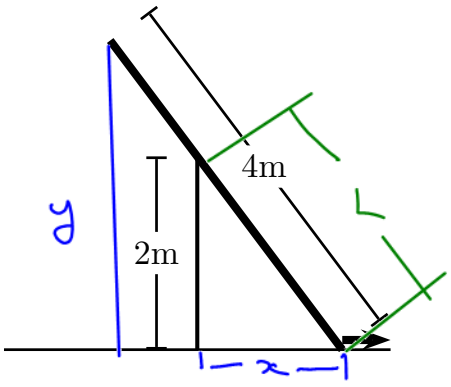
At $(-\sqrt{3},0)$, $\frac{dy}{dx} = \frac{0 - 2(-\sqrt{3})}{2\cdot 0 - (-\sqrt{3})} = \frac{2\sqrt{3}}{\sqrt{3}} = 2$

both slope values are equal.

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8. A 4 m long ladder has one end on the ground, and a point in its middle resting on the top of a 2 m fence. [/6]

If you pull the bottom of the ladder away from the fence at a constant 0.1 m/s, the top of the ladder will move downwards and right.
Focusing specifically on the downwards component of motion, at what speed will the top of the ladder be moving downwards when the ladder's bottom is 2 m away from the bottom of the fence?



Know $\frac{dx}{dt} = 0.1$ m/s,

Want $\frac{dy}{dt}$.

By similar triangles, $\frac{y}{4} = \frac{2}{L}$ ①

but we don't want L in our formula, we want x:

By Pythagoras, $x^2 + 2^2 = L^2$, so $L = \sqrt{4+x^2}$

→ ①: $\frac{y}{4} = \frac{2}{\sqrt{4+x^2}}$ or $\boxed{y = \frac{8}{L}}$

Take $\frac{d}{dt}$ of both sides: $\frac{d}{dt}(y) = \frac{d}{dt}\left(8(4+x^2)^{-1/2}\right)$

$$\frac{dy}{dt} = 8\left(\frac{-1}{2}(4+x^2)^{-3/2} \cdot (2x)\frac{dx}{dt}\right)$$

tidy: $$\frac{dy}{dt} = -8x(4+x^2)^{-3/2} \cdot \frac{dx}{dt}$$

At $x = 2$, $4+x^2 = 8$, so

$$\frac{dy}{dt} = -8(2)(8)^{-3/2}(0.1) \cong -0.0707 \text{ m/s}$$

The tip of the ladder is moving down at 0.0707 m/s.

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Space for additional work. **Indicate clearly which question you are continuing if you use this space.**

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